

Market Successes & Failures

Unit 10 — Study Guide & Lecture Notes

- › Externalities: private vs social cost, and inefficiency
- › Coase bargaining, Pigouvian taxes & compensation
- › Public goods & asymmetric information
- › The limits of markets

BEFORE YOU BEGIN

How to use these notes & where this unit fits

This guide follows the Unit 10 slides section-by-section and expands the reasoning so you can read it alone or annotate it in class. The banana-and-fishery externality diagrams are redrawn cleanly, with the private, external and social cost curves and the efficient quantity computed so the geometry is exact.

DEFINITION BOX

The precise definition to reproduce in the exam.

WORKED EXAMPLE

Numbers carried through, matching the slide examples.

LISTEN FOR IN THE LECTURE

Points usually stressed verbally — leave room for your own notes.

CHECK YOURSELF

Quick questions to test recall.

Where this unit sits

The previous units showed that a **perfectly competitive market generates an efficient (Pareto-optimal) outcome**. But that result relies on strong conditions: **property rights must be well defined**, institutions must be able to **enforce** them, and they must be written into **complete, enforceable contracts**. Often these conditions **fail** — and the market delivers an inefficient outcome. This unit catalogues the main **market failures** and the tools to fix them.

THE FOUR SOURCES OF MARKET FAILURE (ROADMAP)

- **External effects** (externalities) — Section B.
- **Public goods** — Section C.
- **Asymmetric information** — Section C.
- **Market power** ($P > MC$) — Section C.

Section D then asks a deeper question: even where markets *work*, **should** they allocate everything?

Outline

1. **A.** Introduction — when the efficiency result breaks
2. **B.** External effects — inefficiency, bargaining, government intervention
3. **C.** Other types of market failure — public goods, asymmetric information, market power
4. **D.** Limits to markets — should everything have a price?

SECTION A

Introduction

In perfectly competitive markets the market generates an **efficient outcome** — but only if:

- **property rights are well defined**,
- institutions can **enforce** those property rights, and
- they can be written into **complete and enforceable contracts**.

Often these conditions are **not** satisfied — most importantly because of **external effects (externalities)** and **asymmetric information**.

Motivating example — the pesticide chlordane

The **banana weevil** caused large losses on banana plantations. The pesticide **chlordane** was highly effective against it — but **hazardous to humans and the ecosystem** (known since the 1970s). It was used heavily on Caribbean plantations in Guadeloupe and Martinique until 1993:

- The chemical washed off the land into rivers, **contaminating freshwater prawn farms and lobster fisheries**.
- By 2012, Martiniquean men had **the highest rate of prostate cancer in the world**; neurological damage in children was documented.

THE CORE IDEA

The banana plant owners **did not take into account the costs they imposed on fishermen** (and on health). A cost (or benefit) that one party's decision imposes on another, but that the decision-maker ignores, is an **externality** — and it is a source of market failure.

SECTION B

External Effects

DEFINITION — EXTERNALITY

An **external effect (externality)** arises when one party **directly conveys a cost or benefit to another party** that is not reflected in market prices.

- a **cost** imposed on others → **negative externality** (e.g. factory emissions);
- a **benefit** conferred on others → **positive externality** (e.g. a store's advertising helping neighbours).

WHY EXTERNALITIES BREAK EFFICIENCY

Economic efficiency requires **marginal social benefit = marginal social cost**. But an individual firm or consumer only considers its **own (private)** costs and benefits. With externalities, **marginal social cost/benefit differs from marginal private cost/benefit** — so the private optimum is **not** the efficient one. **Maximum efficiency is not achieved.**

Negative externality — bananas and fisheries

Abstracting from the health issues, growing bananas imposes a cost on fisheries (lower catch). Three cost concepts:

THREE MARGINAL COSTS

- **Marginal private cost (MPC)** — the plantation's own marginal cost of growing bananas.
- **Marginal external cost (MEC)** — the extra cost imposed on fishermen per ton of bananas.
- **Marginal social cost (MSC = MPC + MEC)** — the full cost to society.

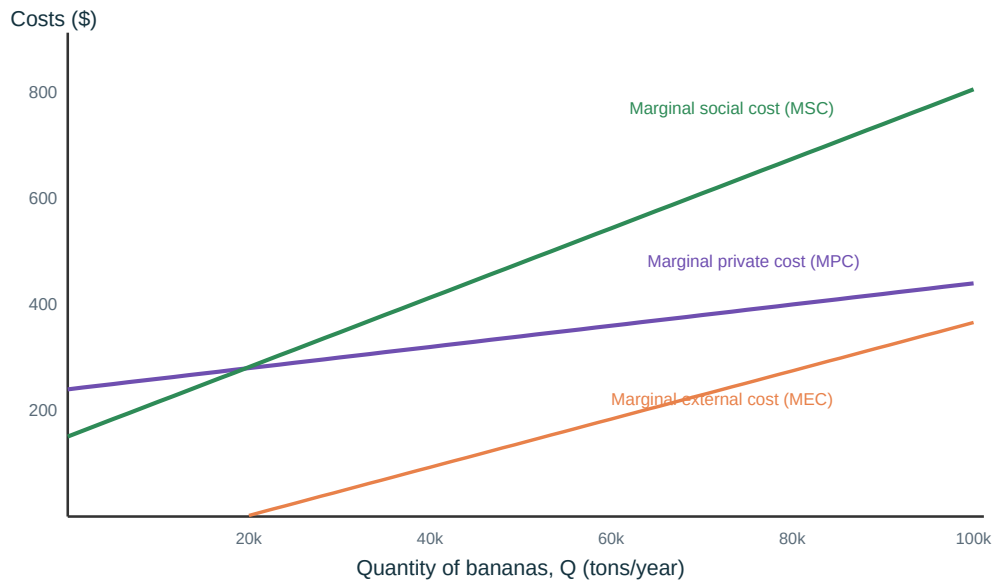


Figure 1. The marginal social cost (green) is the vertical sum of the plantation's marginal private cost (purple) and the marginal external cost imposed on fishermen (orange). The gap between MSC and MPC is exactly the externality.

The private outcome is not efficient

The plantation produces where its **marginal benefit (= the price of bananas)** equals its **marginal PRIVATE cost** — point A at **80,000 tons**. Society, however, should produce where marginal benefit equals marginal **SOCIAL** cost — point B.

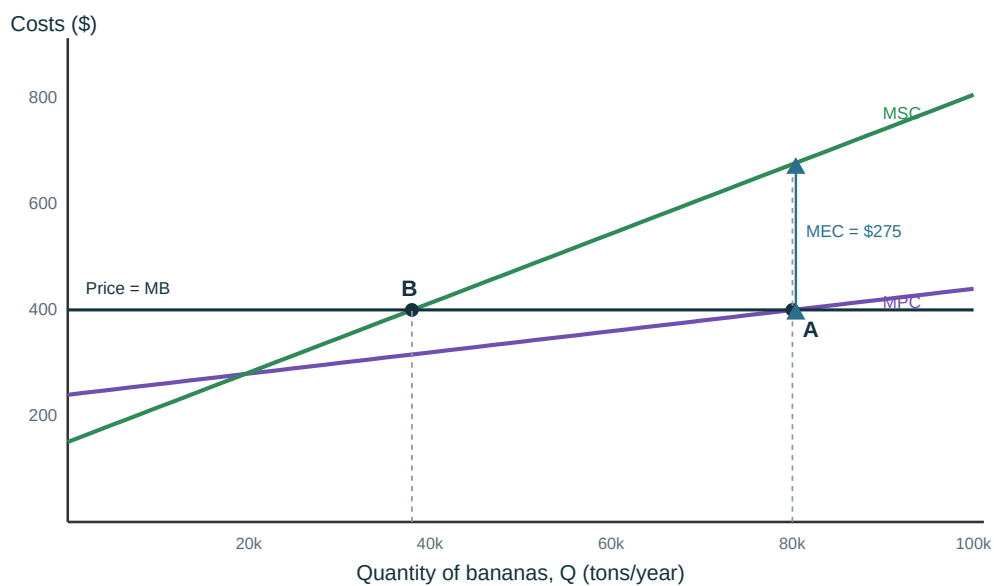


Figure 2. The plantation chooses A (80,000 tons), where price = MPC. But at 80,000 tons the marginal external cost is **\$275** — the gap between MSC and MPC. Since price (marginal benefit) is **below** the marginal social cost there, the last tons are socially wasteful. The outcome is **not Pareto-efficient**.

WORKED EXAMPLE — A PARETO IMPROVEMENT EXISTS

At 80,000 tons the marginal external cost is **\$275**. Fishermen would pay up to **\$275** to cut banana output by one ton. Cutting production by one ton lowers the plantation's revenue by **\$400** (the price) but also lowers its cost by nearly \$400 ($MPC \approx \text{price at A}$), so the plantation would accept **any compensation above roughly zero** to cut back. Since fishermen will pay up to \$275 and the plantation needs almost nothing, **both can be made better off** — the hallmark of an inefficient starting point.

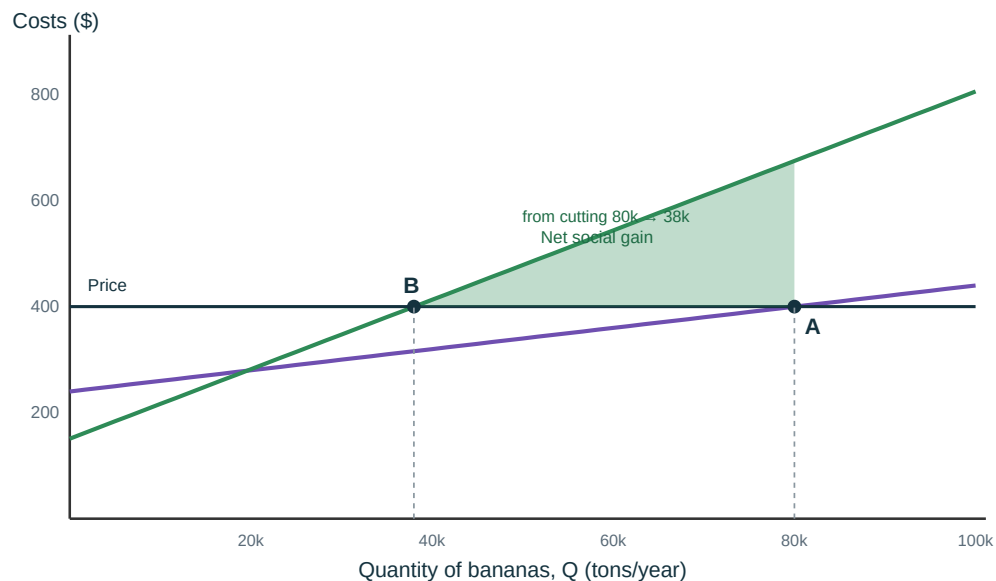


Figure 3. Pareto improvements are possible as long as the price (marginal benefit) is **below** the marginal social cost. The efficient outcome is where **marginal benefit = MSC** — a production rate of about **38,000 tons**. The green triangle is the **net social gain** from cutting production from 80,000 to 38,000.

Positive external effects

Actions can also confer **benefits** on others: a **positive externality** where the **marginal social benefit exceeds the marginal private benefit**. Example: a large store (e.g. FNAC) opening in a shopping street also benefits **other shops selling non-competing products**. Positive externalities lead to **under-provision** (too little of the good activity), the mirror image of the over-production under negative externalities.

LISTEN FOR IN THE LECTURE

The whole section rests on one move: replace the **private** cost/benefit the decision-maker sees with the **social** one. Negative externality → private cost too low → **too much** produced. Positive externality → private benefit too low → **too little**. Every policy below is a way to make the decision-maker face the social figure.

Fixing Externalities

Solution #1 — Bargaining (the Coase idea)

COASE THEOREM

Legally **assign property rights** over the externality (e.g. the right to pollute, or the right to clean water). Then **private bargaining** between the parties will reach a **Pareto-efficient allocation regardless of which party holds the rights** — in the absence of transaction costs. Private bargaining can be **more effective than government intervention** because the parties have more of the relevant **information**.

But **transaction costs** — the costs of acquiring information, enforcing the contract, or organising collective action — can be a major obstacle in reality.

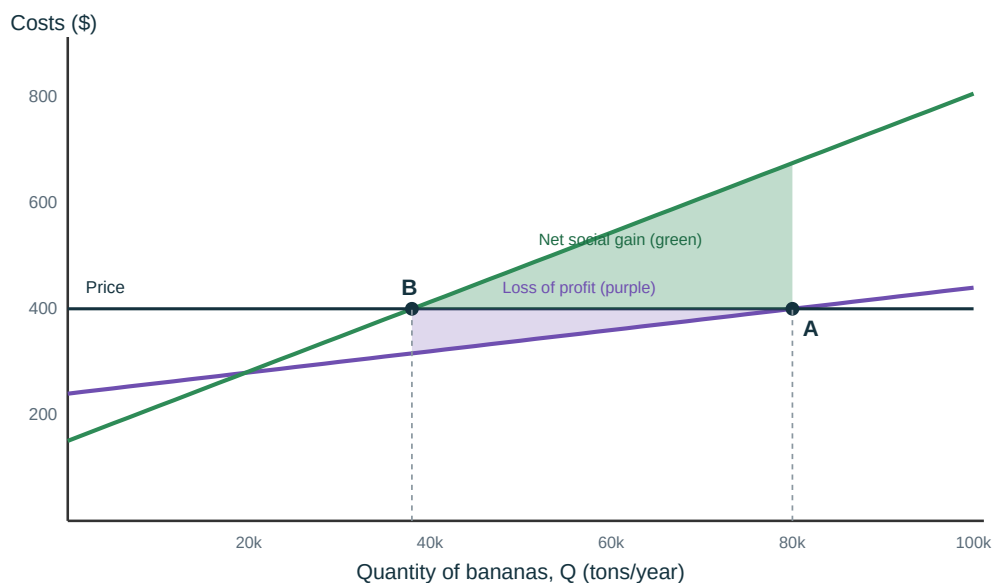


Figure 4. The size of the deal. The **plantation's minimum acceptable offer** = its **lost profit** from cutting output (purple area between price and MPC over 38k–80k). The **fishermen's reservation option** (maximum they'd pay) = the **whole external cost avoided** = green + purple. Because the **net social gain** is positive, a deal exists; where in the range the price lands depends on **relative bargaining power**.

PRACTICAL LIMITS OF BARGAINING

- **Collective action** — finding a representative and agreeing how to split the gains within each side.
- **Missing information** — calculating each fisherman's exact loss and each plantation's contribution to pollution.
- **Enforcement** — a court may struggle to verify whether plantations complied.
- **Limited funds** — fishermen may not have the money to pay the required compensation.

Solution #2 — Government intervention

Three instruments, each with **different distributional consequences** (and each with limits: missing information / measuring social costs, and lobbying).

(a) Regulation

The government **caps** banana production at the Pareto-efficient level (38,000 tons). Plantation profits fall (lower volume), but there is no tax bill.

(b) Pigouvian taxes / subsidies

DEFINITION — PIGOUVIAN TAX/SUBSIDY

A tax on firms generating a **negative** externality (or a subsidy for a **positive** one), set **equal to the marginal external cost at the Pareto-efficient level**. It **forces producers to face the full social cost** of their decisions, so their private optimum coincides with the social optimum.

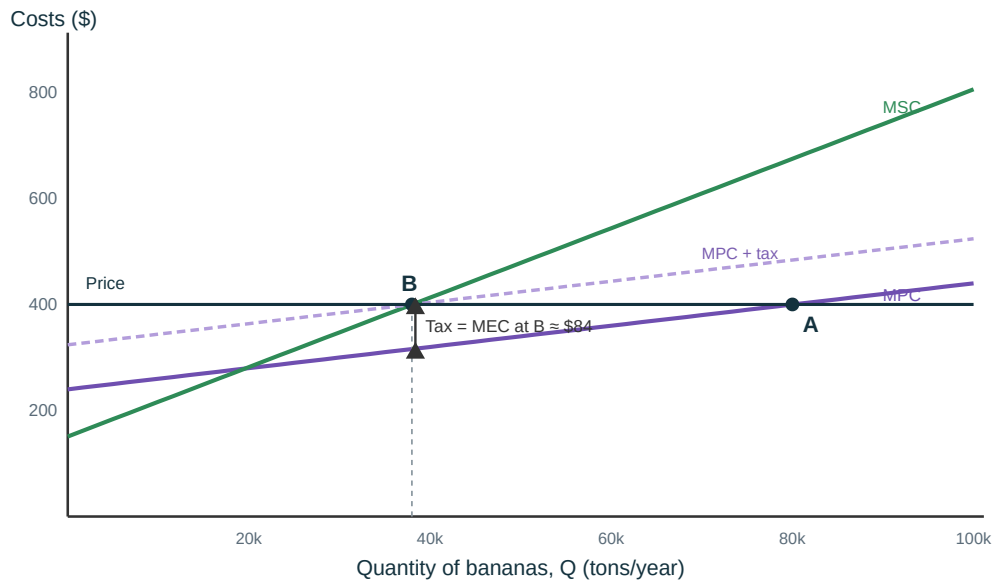


Figure 5. A Pigouvian tax ($\approx$ the marginal external cost at B) shifts the plantation's effective marginal cost up to **MPC + tax** (dashed). Its new private optimum is now at **38,000 tons** — the efficient level. Note this is **worse for plantations than regulation**: they cut volume *and* pay the tax.

(c) Compensation

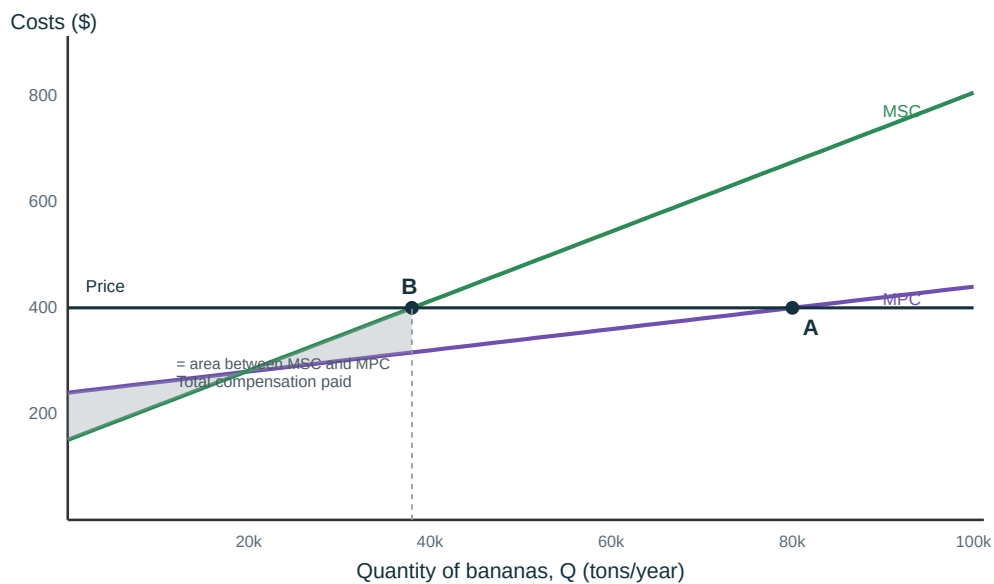


Figure 6. Plantations are required to **compensate** fishermen by the difference between marginal social and marginal private cost (= the marginal external cost) for each ton produced. This drives them to the efficient 38,000 tons. Compared with regulation, plantation profits are **lower** but **fishermen are better off** (they receive the compensation).

Why do external costs occur? Incomplete contracts

ROOT CAUSE

External costs cause market failure because of **incomplete contracts** — contracts that do not specify, in an **enforceable** way, every aspect of the exchange affecting all parties. Contracts that would internalise external costs are unenforceable because the relevant information is **not verifiable** or is **asymmetric** (not known to the decision-maker). So in reality it is **impossible** to use contracts or property rights to fold all social costs into the decision.

REAL-WORLD TWIST — DISNEY'S SECRET LAND PURCHASES

Disney bought the land **around** its Anaheim park **secretly** (and in Orlando via multiple shell companies / agents; in Hong Kong via an **option** on adjoining real estate). Why secretly? Because a park generates a huge **positive externality** on surrounding land values. If sellers knew Disney was buying, they would hold out for the higher post-park price — so Disney bought quietly to **internalise the externality** (capture the land-value gain) itself.

SECTION C

Other Types of Market Failure

Besides externalities, markets fail to reach a Pareto-efficient outcome for three more reasons: **public goods**, **asymmetric information**, and **market power**.

Public goods

Goods are classified along two dimensions — whether they are **rival** (my use reduces yours) and whether they are **excludable** (I can stop non-payers using them).

DEFINITIONS

- **Non-rival** — use by one person **does not reduce** its availability to others (a flood-control dam, street lighting, national defence, a TV broadcast).
- **Non-excludable** — it is **impossible to exclude** anyone from access (fish in the open ocean, flood protection).
- A **public good** is a **non-rival** good (which may or may not be excludable).

	Rival	Non-rival
Excludable	Private goods (food, clothes, houses)	Artificially scarce goods (subscription TV, uncongested toll roads, patented knowledge)
Non-excludable	Common-pool resources (ocean fish stocks, common grazing land)	Public goods & "bads" (a lunar eclipse view, public broadcasts, national defence, noise & air pollution)

After CORE Figure 12.8 — private goods and public goods.

WHY MARKETS FAIL FOR PUBLIC GOODS

Markets allocate **private** goods well, but tend to fail for the other three types:

- **Non-rival goods** have a **marginal cost of zero**, so pricing at $P = MC$ is impossible unless provision is **subsidised** or there is **market power**.
- **Non-excludable goods** can't be priced at all — the provider **cannot exclude non-payers** (the **free-rider** problem). Example: if one farmer builds a community irrigation project, others enjoy the **external benefit** for free, so it is hard to make them pay or to write a contract guaranteeing the efficient level.

Asymmetric information

DEFINITION

Information is **asymmetric** when **one party knows something relevant to the transaction that the other does not**. Two forms:

- **Hidden action** → a **moral hazard** problem (an action one party takes that the other can't observe). *Example: insured people take more risks.*
- **Hidden attributes** → an **adverse selection** problem (a characteristic one party can't observe). *Example: used-car buyers don't know the car's quality, but sellers do.*

ADVERSE SELECTION — HEALTH INSURANCE

Insurance is voluntary and the insurer **can't observe** buyers' health; buyers **know** theirs, and the less healthy are keener to buy. To break even the insurer must charge a **high price**, so **only the less healthy buy** — most buyers already have a problem. The result is a **missing market**: many healthier people who would like insurance stay uninsured.

MORAL HAZARD — CAR INSURANCE

Any insurance also has a **hidden-action** problem: once covered, the buyer takes **more risk** (full damage cover → more careless driving). Insurers can write **some** contractual limits but **cannot enforce** others (e.g. your driving speed), so the behaviour can't be fully controlled.

Market power — price above marginal cost

Firms with **market power** set **price above marginal cost**. This is a **market failure** because the allocation is **not Pareto-efficient** (some consumers who value the good above its marginal cost don't buy it). Policies try to **limit market power** — but with two cautions:

- **Incentives for innovation** — some markup can be the reward that funds R&D.
- Market power arising **solely from product differentiation** may be fine — people **value having different choices**.

SECTION D

Limits to Markets

Everything so far asked *when markets fail to be efficient*. A different, deeper question: even where a market **would** function, **should** it allocate the good at all?

ARGUMENTS AGAINST USING MARKETS FOR EVERYTHING

- **Repugnant markets** — creating a market for some goods/services would **violate ethical or social norms** (e.g. a market in people — slavery).
- **Other institutions may be more effective** — governments, families.
- **Crowding out** — market mechanisms can **crowd out** norms and social preferences (paying for something can destroy the intrinsic motivation to do it).
- **Merit goods** — goods that **should be available to everyone regardless of ability to pay** (e.g. education, basic health care).

LISTEN FOR IN THE LECTURE

This section is **normative**, not just positive: it's about values, not only efficiency. A market can be perfectly efficient *and* still be the wrong way to allocate something. Keep the two questions separate in the exam — "is it efficient?" vs "should it be a market at all?"

Summary, Formulas & Glossary

The unit in one paragraph

Competitive markets are efficient only when property rights are complete and enforceable. They fail through **externalities** (decisions impose uncoded effects on others — $MSC \neq MPC$, so the private optimum over- or under-produces), **public goods** (non-rival goods have zero marginal cost; non-excludable goods invite free-riding), **asymmetric information** (hidden actions → moral hazard; hidden attributes → adverse selection and missing markets), and **market power** ($P > MC$). Externalities can be tackled by **bargaining** (Coase — efficient regardless of who holds the rights, absent transaction costs), or by government **regulation**, **Pigouvian taxes/subsidies**, or **compensation** — instruments that reach the same efficient quantity but differ in **who bears the cost**. Finally, some goods arguably **should not** be allocated by markets at all (repugnant markets, crowding out, merit goods).

Concept sheet

Concept	Meaning / rule
Efficiency condition	marginal social benefit = marginal social cost
MSC	$MSC = MPC + MEC$ (private + external marginal cost)
Private optimum (neg. externality)	price = MPC → over-production
Efficient quantity	price = MSC
Coase theorem	clear property rights + zero transaction costs → efficient bargain, whoever holds the rights
Pigouvian tax	tax = MEC at the efficient quantity
Public good	non-rival ($\pm$ excludable); $MC \approx 0$
Moral hazard	hidden action (unobservable behaviour)
Adverse selection	hidden attributes (unobservable type) → missing market
Market power failure	$P > MC$

Glossary

Term	Meaning
Externality	Cost/benefit a decision imposes on others, ignored by the decision-maker.
MPC / MEC / MSC	Marginal private / external / social cost.
Pareto efficiency	No one can be made better off without making another worse off.
Coase theorem	With clear rights and no transaction costs, bargaining reaches efficiency regardless of who holds the rights.
Transaction costs	Costs of information, enforcement, collective action.
Pigouvian tax/subsidy	Tax/subsidy set to the marginal externality at the efficient level.
Incomplete contract	Contract that doesn't enforceably cover all affected interests.
Rival / excludable	My use reduces yours / non-payers can be shut out.
Public good / common-pool resource	Non-rival ($\pm$ excludable) / rival but non-excludable.
Free-rider problem	Enjoying a non-excludable good without paying.
Moral hazard / adverse selection	Hidden action / hidden attributes under asymmetric information.
Missing market	A market that fails to exist because of adverse selection.
Merit good / repugnant market	Should be available to all / a market that violates social norms.

Exam-style questions to practise

1. A factory's $MPC = 10 + 0.5Q$ and it imposes $MEC = 0.5Q$ on neighbours; the price is 40. Find the private and the socially efficient output, and the size of the deadweight loss.
2. Explain the Coase theorem. Why does the efficient outcome not depend on whether the polluter or the victim holds the property right? When does it break down?
3. Compare regulation, a Pigouvian tax and compensation for the banana example: which reaches the efficient quantity, and how do they differ in distribution?
4. Classify each as rival/excludable: a lighthouse, a congested motorway, a patented drug, an ocean fishery. Which are public goods, and why might markets under-provide them?
5. Distinguish moral hazard from adverse selection with an insurance example of each. How does adverse selection produce a "missing market"?
6. Give two goods you think should *not* be allocated by markets, and justify using the "limits to markets" arguments.

Study guide prepared as a companion to *Microeconomics — Prosperity and Inequality, Unit 10: Market Successes and Failures* (KU Leuven, 2025–2026). The banana/fishery curves are computed so the private optimum (80,000 tons, MEC = \$275) and efficient quantity ($\approx 38,000$ tons) match the lecture figures; the public-goods matrix follows CORE Figure 12.8.